INCH-POUND

MIL-P-70735 (AR)
28 May 1991

MILITARY SPECIFICATIONS

PAPER, NITROCELLULOSE BASE FOR ARTILLERY PROPELLING CHARGES

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and packaging for the paper used to manufacture combustible case components for the artillery propelling charge designated as M203A1 Charge, Propelling 155MM.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specification, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-B-117	- Bags, Sleeves and Tubing
MIL-P-46078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-EAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1905	- Design and Test Requirement for Level B and Level C Ammunition Packaging

(Unless otherwise indicated, copies of the federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9345109	- Composition
9357034	- Composition

(Copies of drawings, publications and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by contracting officer.)

CODE OF FEDERAL REGULATIONS

49 CFR	- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(Application for copies should be addressed to the Superintendant of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR latest revision)).

2.2 Non-Government publications. The following documents forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in

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the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI T410	Weight Per Unit Area (Basic Weight or Substance) of Paper and Paperboard.
TAPPI T411	Thickness (Caliper) of Paper and Paperboard.
TAPPI T494	Tensile Breaking Properties of Paper and Paperboard (Using Constant Rate of Elongation Apparatus).

(Application for copies should be addressed to: Technical Association of the Pulp and Paper Industry, TAPPI, 1 Dunwoody Park, Atlanta, GA 30341).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D685	Conditioning paper and paper products for testing.

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.1.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Paper. The paper shall comply with all the requirements specified in this specification, and with all requirements specified in applicable specifications and compositions listed on drawing 9345109 and 9357034 when tested in accordance with 4.5.1.

3.2 Form and dimensions. The paper shall be furnished in rolls of the width and length as specified in the contract or order.

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3.3 Physical properties. The basis weight, thickness (caliper), tensile strength, tearing resistance and bursting strength of the paper shall meet the requirements of Table I.

Table I

Basis weight, thickness (caliper), tensile strength, tearing resistance and bursting strength requirements (see Note 1)

Physical properties (Single Sheet)		52% NC (see Note 2)	52% NC (see Note 3)	72% NC (see Note 4)
Basis Weight	Maximum	465g/m^2	615g/m^2	535g/m^2
	Minimum	435g/m^2	585g/m^2	505g/m^2
Thickness (Caliper)	Maximum	0.50 MM	0.85 MM	0.74 MM
	Minimum	0.40 MM	0.75 MM	0.65 MM
Tensile Strength	(See Note 5)	110 daN/cm^2	65 daN/cm^2	45 daN/cm^2
	(See Note 6)	90 daN/cm^2	60 daN/cm^2	30 daN/cm^2
Moisture Content	Maximum	16%	16%	16%
	Minimum	2%	2%	2%

Note 1: Any deviations from the physical property requirements will require approval from the technical agency (see 6.6)
Note 2: For 5 ply product
Note 3: For 4 ply product
Note 4: For 4 ply product
Note 5: Machine direction
Note 6: Cross direction

3.4 Stability. The nitrocellulose in the paper shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested in accordance with 4.5.2.

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3.5 Percent by weight of resin, additives, and kraft. The percent by weight of resin, additives, and kraft shall be determined when tested in accordance with 4.5.1.3.5.

3.6 Workmanship.

3.6.1 Rolls before wrapping. The paper shall have no lumps, spots or high spots before being made into a roll. The rolls of paper shall be free from dirt, grit, oil, grease and other foreign material. The paper shall have no blue, gray or other nonconforming colored spots.

3.6.2. Rolls after wrapping. The rolls of paper and wrapping shall be free from oil, grease and other foreign material that may be absorbed by the wrapping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may used his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Goverment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contact. Sampling inspection, as parts of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article inspection (4.3)

b. Quality conformance inspection (4.4)

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated in Table I:

Table I. First Article Sample

Part Description	Drawing	Quantity
Paper, Nitrocellulose Base	9345109	5
Paper, Nitrocellulose Base	9357034	5

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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Table II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70735 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Paper, Nitrocellulose Base (Dwg. 9345109)	5	3.2	4.4.2.1
	Paper, Nitrocellulose Base (Dwg. 9357034)	5	3.2	4.4.2.2

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

Each lot of paper shall contain:

a. Nitrocellulose from one manufacturer and from not more than two lot numbers with the same interfix number.

b. Diphenylamine from one manufacturer and from not more than two lot numbers with the same interfix number.

c. Resin from one manufacturer and from not more than two lot numbers with the same interfix number.

d. Kraft from one manufacturer and not more than two lot numbers with the same interfix number.

e. Talc from one manufacturer and not more than two lot numbers with the same interfix number.

4.4.2 Examinations and tests. See MIL-A-48078.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited in the classification of characterisitcs paragraphs, attributes sampling inspection shall be conducted in accordance with Table III below.

Table III

Attributes Sampling for Major and Minor Defects

Lot Size (Rolls/lot)	Major Sample Size See Note 1	Minor Sample Size See Note 1
2 to 8	*	See Note 2
9 to 15	*	5
16 to 25	*	5
26 to 50	32	5
51 to 90	32	12

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Table III

Sampling Plan for Major and Minor Defects

Lot Size (Rolls/lot)	Major Sample Size	Minor Sample Size
91 to 150	32	12
151 to 280	32	14
281 to 500	32	17
501 to 1200	74	20
1201 to 3200	74	24
3201 to 10000	87	30
10001 to 35000	109	36
35001 to 150000	124	40
150001 to 500000	156	40
500001 and over	187	40

[a] 100% Inspection

NOTE 1: Sampled rolls shall be randomly withdrawn from the lot. One sample shall be taken from each sampled roll. Samples shall be taken from different locations of the rolls, i.e., the right side, the middle or the left side of the roll.

NOTE 2: Use 100% inspection when lot size is less than 5 rolls of nitrocellulose paper. Use sample size of 5 rolls when lot size is greater than 5 rolls of nitrocellulose paper.

b. Alternative quality conformance provisions. Alternative quality conformance provisions, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70735 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Paper, Nitrocellulose Base	1 OF 1	9345109/9357034
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Tensile strength test	Table III	3.3	4.5.3.
102	Thickness	Table III	3.3	4.5.4
103	Basis weight	Table III	3.3	4.5.5
104	Stability	4.4.3.2	3.4	4.5.2
105	Composition	4.4.3.1	3.1	4.5.1
106	Moisture	Table III	3.3	4.5.6
Minor				
201	Evidence of poor workmanship	Table III	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70735 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Paper, Nitrocellulose Base, After Wrapping	1 OF 1	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Wrapping material torn or not secure	100%	3.6.2	Visual
Minor 102	Evidence of poor workmanship	Table III	3.6	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4.3 Testing.

4.4.3.1 Composition (see dwg. 9345109 and 9357034). To determine the composition of the nitrocellulose paper, one felting sample shall be withheld from the beginning, middle and end of each slurry batch production of the lot. From each such sample, the composition shall be tested in accordance with 4.5.1. Failure of any specimen is cause for rejection of all rolls manufactured from that batch since the last previously acceptable analysis and until an acceptable analysis is performed on a subsequently manufactured sample from that batch.

4.4.3.2 Stability. Five samples shall be selected from each lot and tested in accordance with 4.5.2. The lot shall be rejected if any sample fails to comply with the requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Composition.

4.5.1.1 Preparation of samples. Each sample shall be cut into approximately 1/4 inch squares. From each sample, approximately 0.5 gram shall be selected for testing and weighed to the nearest 0.2 milligram (Weight A) in an extraction thimble. The same thimble shall be utilized during the extractions in 4.5.1.3.1 and 4.5.1.3.3.1.

4.5.1.2. Total volatiles. The samples from 4.5.1.1 shall be dried for 3 hours at 100 ± 2 Degrees C, cooled in a desiccator (1/2 hr. min.), and weighed to the nearest 0.2 mg (Weight W). Calculate the percent total volatiles as follows:

$$\text{Percent (\%) Total Volatiles} = \frac{A - W}{A} \times 100$$

4.5.1.3 Analytical procedure.

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4.5.1.3.1 Solvent extraction. Using a tared, extra-coarse, sintered glass thimble, extract each sample from 4.5.1.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286, Method 104.1.3). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.1.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 2 Degrees C. Cool in a desiccator (1/2 hr. min.). Determine the dry weight of this residue (Weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. The residue is to be used for the nitrocellulose determination in 4.5.1.3.3.1.

4.5.1.3.2 Diphenylamine (DPA). To the dried extract from 4.5.1.3.1 add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1.4 (starting with paragraph 5.5) of MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in MIL-STD-286, Method 210.1.4, paragraph 5.13.

4.5.1.3.3 Nitrocellulose - Schulze - Tiemannn Method for Nitrocellulose (NC) determinations.

Assemble the 250 ml Kjeldahl flask, gas evolution apparatus eudio-meter/condenser tube, NaOH bath, lead band, heating mantle, transformer and protective shield as shown in Figure I.

Calibrate the apparatus with dry standard A.C.S. grade KNO_3. Weigh 0.5 - 0.7 g standard samples to 0.1 mg and analyze using the following procedure. The factor is determined using the following equation:

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$$F = \frac{616.8\ W\ (273.15 + T)}{V\ (B - C - P - S)}$$

F = Eudiometer Factor
W = Mass of KNO3 (g)
T = Temperature of Condenser Water (Degrees C)
V = Volume of Gas in Eudiometer Tube (ml)
B = Barometric Pressure (mm Hg), Fig. 2
C = Temperature Correction (mm Hg) Table IV
P = Vapor Pressure of Water over 33% w/w NaOH at T (mm Hg) Table V
S = Local Gravity & Sea Level Values corrected to standard values

Prepare samples as follows: Cut a speciman into 1/4 inch or smaller squares. Dry to constant weight using microwave moisture analyzer.

Displace the nitric oxide (NO) from the previous run by aspirating the gases through the drain or venting through a hood. Open the stopcock at the top of the eudiometer tube and carefully raise the leveling bulb until all the gas has been expelled. Rest the bulb in its holder and stopper it. Loosen the eudiometer tube from the rubber stopper in the bottom of the bath and drain the NaOH in the tube into the tank. Fill the leveling bulb to 90% of its capacity and place the empty eudiometer tube back onto the rubber stopper and secure it. Fill the eudiometer to the 0.0 ml mark with the leveling bulb, removing all air bubbles, close the stopcock, secure the bulb and stopper it. Replace the 33% w/w NaOH as needed to maintain a constant level in the tank.

Quantitively transfer 0.8 - 1.0 g (weighed to $\pm$ 0.1 mg) of the dry sample into a 250 ml Kjeldahl flask, add 50 mL deionized water and a few glass beads to prevent bumping. Attach the addition funnel/check valve assembly to the freshly greased joint of the Kjeldahl flask. Attach spring clamps or rubber bands to the ears of the flask and addition funnel. Adjust the gas evolution assembly so that the exit tube is gently resting on the bottom of the NaOH bath. Place the lead band round the exit tube opening.

Open the addition funnel stopcock and turn on the heating mantle. Following the purging of the Kjeldahl flask, close the addition funnel stopcock and purge the rest of the tubing for 5-10 minutes. The system is thoroughly purged when no more bubbles escape into the bath.

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Remove the heating mantle from the Kjeldahl flask and observe the drawing of the NaOH solution into the exit tubing and the closing of the check valve. Remove the eudiometer tube from the rubber stopper, place it over the open end of the exit tube. Rest the bottom of the eudiometer tube on the lead band and secure the eudiometer. Refill the leveling bulb to 90% of its volume, stopper the bulb and place it back into its ring.

Place 25 ml of FeCl2 solution into the addition funnel. Open the stopcock slowly and allow most of the FeC12 to flow into the flask. Retain a few drops in the stopcock to preclude the flow of air into the system. If air is allowed into the system, the analysis is invalid and must be repeated. Place 25 ml of concentracted HCl to the addition funnel and add it to the Kjeldahl flask. Add 10 mL deionized water in a similar fashion. Following the last addition, fill the addition funnel with 30 ml deionized water and tighten the stopcock nut. Place the heating mantle around Kjeldahl flask and heat, for 20-30 minutes or until bubbling has stopped. Remove the heating mantle and check to see that the valve closes. Loosen the stopcock nut on the addition funnel, allow the water to drip into the Kjeldahl, then increase the flow to effectively rinse the walls of the flask, close stopcock before air enters and tighten the nut. Attach the heating mantle, tap the check valve lightly until it opens and continue for 5-10 more minutes.

Following the completion of the reaction, remove the heating mantle and observe the closing of the check valve. Carefully move the eudiometer tube from its present position back to the rubber stopper and secure it. Slowly release the vacuum inside the Kjeldahl flask by allowing water and air to enter the flask through the stopcock.

Tap the check valve to help it open. Allow the gas to equilibrate thermally with the water in the jacket around the eudiometer tube. Wait at least 20 minutes. Measure the volume of gas collected in the eudiometer tube by uncorking the leveling bulb and adjusting it so that the top of the fluid in the bulb is at exactly the same height as the top of the fluid in the eudiometer tube. Read the volume to the nearest 0.1 ml. Use the following equation to calculate the % NC in the sample:

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$$\% \text{ NC} = \frac{2.246\ V\ (B - C - P - S)\ F}{NW\ (273.15 + T)}$$

V = Volume of Gas in Eudiometer Tube (ml)
B = Barometric Pressure (mm Hg)
C = Temperature Correction (mm Hg) Table II
P = Vapor Pressure of Water over 33% (w/w) NaOH at T (mm Hg) Table III)
F = Eudiometer Factor
N = Percent Nitrogen in NC (%)
W = Mass of Sample (g)
T = Temperature of Condenser Water (Degrees C)
S = Local Gravity & Sea Level values corrected to standard values

PREPARATION OF FeCl2:

Prepare the solution by adding 1000 g of FeCl2.4H2O, 1200 ml deionized water and 70 ml of concentrated HCl to a 5 liter round bottom flask. Attach the reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron filings or powder to the cool solution. When the gas evolution has stopped, filter quickly without aerating the solution to remove the fines. Use A.C.S. grade chemicals throughout the synthesis.

Prepare the bath fluid by dissolving reagent grade NaOH in deionized water until the concentration is 33% weight to weight.

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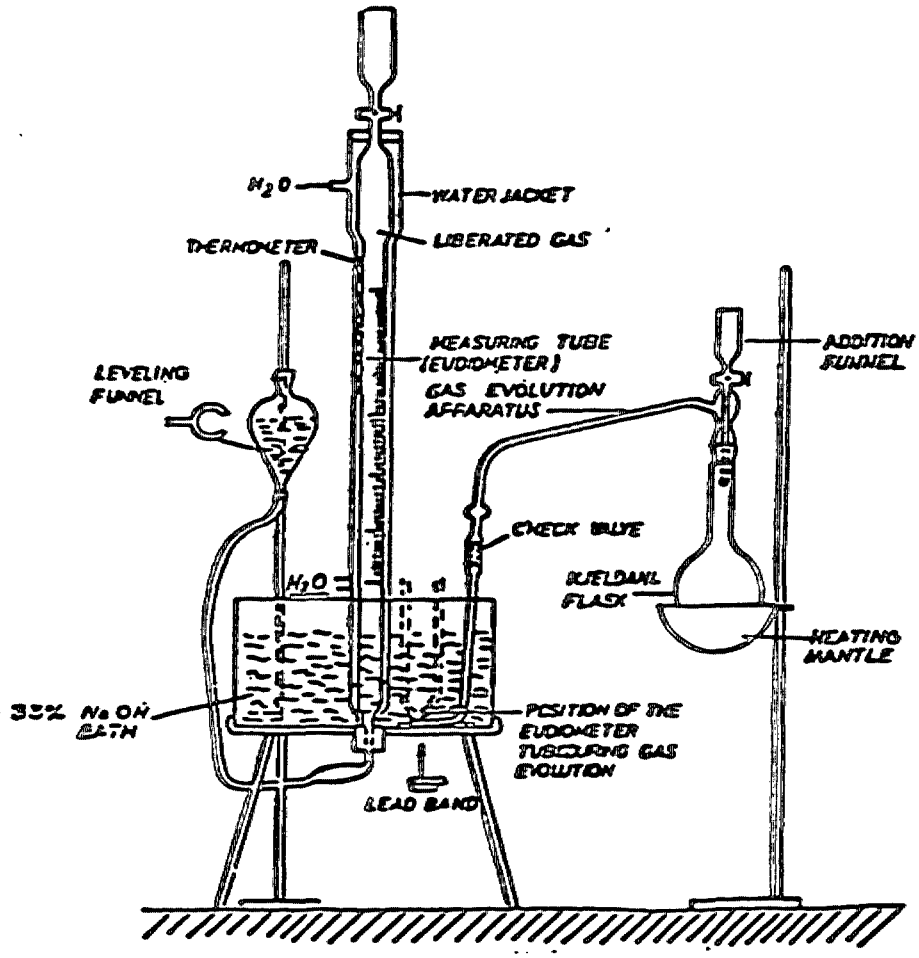


FIGURE 1.

SCHULTZ - TIEMANN APPARATUS

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TABLE IV

	BAROMETRIC PRESSURE IN mm Hg		
	740	760	780
Temp (See Note 1)			
14	1.69	1.73	1.78
15	1.81	1.86	1.91
16	1.93	1.98	2.03
17	2.05	2.10	2.16
18	2.17	2.23	2.29
19	2.29	2.35	2.41
20	2.41	2.47	2.54
21	2.53	2.60	2.67
22	2.65	2.72	2.79
23	2.77	2.84	2.92
24	2.89	2.97	3.05
25	3.01	3.09	3.17
26	3.13	3.21	3.30
27	3.25	3.34	3.42
28	3.37	3.46	3.55

Note 1: Temperature (Degrees C) on Barometer Thermometer.

Vapor Pressure of Water Dyer 33% (w/w) NaOH
Variable F

Temperature of Condenser Water (^{o}C)	18	19	20	21	22	23	24	25
P =	11.6	12.4	13.2	14.0	14.9	15.8	16.8	17.8

Barometric pressure temperature correction, Variable C

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TABLE V

Pressure correction value due to the vapor pressure of sodium hydroxide solution in the eudiometer tube.

TEMPERATURE °C	VAPOR PRESSURE OF NaOH SOLUTION (mmHg)
18	11.6
19	12.4
20	13.2
21	14.0
22	14.9
23	15.8
24	16.8
25	17.8

Calculation of P_a (actual pressure) given a temperature T_a (actual temperature):

$$(T_a - T_1) \times (P_2 - P_1) + P_1 = P_a$$

where: T_a = actual condenser water temperature

T_1 = next lowest tabulated temperature value from T_a

P_2 = pressure at next highest tabulated temp from T_a

P_1 = pressure at T_1

4.5.1.3.3.1 Nitrocellulose (Alternative method). To provide heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and on the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.5.1.3.1 and extract with acetone using the method in 4.5.1.3.1. After extraction dry the residue remaining @ 100 ± 2 Degrees C for one hour. Cool in a desiccator (1/2 hr. min.), weigh to the nearest 0.2 mg (Weight D). Evaporate the acetone extract to dryness and add 25 ml of n-butyl acetate and 45 ml of glacial acetic acid. Replace the air with a stream of inert gas (CO2 or N2). Maintain the inert atmosphere throughout this test. Reflux the solution for five

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minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1.1) and 10 ml concentrated hydrochloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the speciman is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammonium thiocyanate solution (MIL-STD-286, Method 601.1). Titrate with a 0.2N titanous chloride solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate percent NC as follows:

$$\text{Percent (\%) NC} = \frac{(46.69)(N)(V-B)}{(EXW)} \times 100$$

Where:

V = Volume titanous chloride used for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.1.3.4 Talc content (ash test). The specimen of approximately 1 - 1.5 grams shall be taken from the same case body for the above testing and weighed to within 0.2 mg. in a tared porcelain crucible. Cover the crucible. The specimen shall then be ignited at 500 Degrees C in a muffle furnace. The crucible shall be cooled to room temperature and then weighed. Heat until all evidence of carbon is gone (no black residue), or for one hour. The increase in weight of the crucible shall be calculated on a total volatile - free basis to the percent of talc content.

4.5.1.3.5 Resin, Additives and Kraft. Calculate the percent (%) resin, additives and kraft as follows:

% Resin + % additives + % kraft = 100 - (% NC + % DPA + % Talc)

4.5.1.3.6 Methylene chloride extractables. Calculate the total methylene chloride extractables $\frac{(W-C)}{W} \times 100$ and the total acetone extractables $\frac{(C-D)}{W} \times 100$

4.5.2 Stability. Samples shall be tested in accordance with MIL-STD-286, Method 404.1.2 heat tests at 134.5 Degrees C.

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4.5.3 Tensile strength. Each sample shall be tested in accordance with TAPPI-T494 or approved equivalent. Failure of any sample to comply with the applicable requirement shall cause the lot to be rejected.

4.5.4 Thickness. Each sample shall be tested in accordance with TAPPI-T411. Failure of any sample to comply with the applicable requirement shall cause the lot to be rejected.

4.5.5 Basis weight. Each sample shall be tested in accordance with TAPPI-T410. Failure of any sample to comply with the applicable requirement shall cause the lot to be rejected.

4.5.6 Moisture. Each sample shall be tested in accordance with TAPPI-T412. Failure of any sample to comply with the applicable requirement shall cause the lot to be rejected.

5. PACKAGING

5.1 Packaging.

5.1.1 Level A. To be determined.

5.1.2 Level B. To be determined.

5.1.3 Level C. A single Paper, Nitrocellulose Roll whose net weight does not exceed 160 pounds shall be preserved in accordance with Methods IA-8 (Bag, Sealed), of MIL-P-116. Specifically, a bag meeting the requirements of MIL-B-117 Type I, Class F, Style 1, and having a minimum thickness of .004 inches shall be utilized.

5.2 Packing.

5.2.1 Level A. To be determined.

5.2.2 Level B. To be determined.

5.2.3 Level C. Packing shall be in accordance with MIL-STD-1905. A limit of 160 pounds net weight shall apply.

5.3 Marking.

5.3.1 Level A. To be determined.

5.3.2 Level B. To be determined.

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5.3.3 Level C. Mark in accordance with MIL-STD-129, MIL-STD-1905 and 49 CFR Code of Federal Regulations - Transportation, Parts 100-199.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The materials covered by this specification are intended for use in the manufacture of combustible case components used for propelling charges.

6.2 Acquisition requirements. Acquisition documents must specify the following (see MIL-A-48078):

a. Title, number, and date of the specification.

b. Issues of DoDISS to be cited in the solicitation and if required, the specific issue of the individual documents referenced (see 2.1).

c. Provisions for submission of first article sample.

6.3 Data requirements. Deliverable data required by this specification and not referenced elsewhere are cited below. Data shall be furnished to AMSMC-QAR-R (D), ARDEC, Picatinny Arsenal, NJ 07806-5000 and parts and propelling charge assembly manufacturer.

Paragraph	Data Requirement	Applicable DID
3.3, 3.4, 3.5 3.6, 3.7, 3.8	Test and Demonstration Reports	DI-T-1906

6.4 Submission of inspection equipment designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Additives. Additives shall have prior approval of the contracting officer. Processing aids are considered additives.

6.6 Physical properties. The contractor shall submit a description of the deviation from the physical property requirements and the need/reason of the deviation to the Contracting Officer with a copy of submission forwarded to AMSMC-QAR-R (D) for review and approval.

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6.7 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation and approval by the technical activity responsible for preparation of this specification.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading Armament Research Development and Engineer Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.9 Subject term (key word) listing.

Basis weight

Thickness

Tensile strength

Moisture content

Combustible case components

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1375-A466)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70735 (AR)	28 May 1991

3. DOCUMENT TITLE: PAPER, NITROCELLULOSE BASE FOR ARTILLERY PROPELLING CHARGES

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	(IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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